

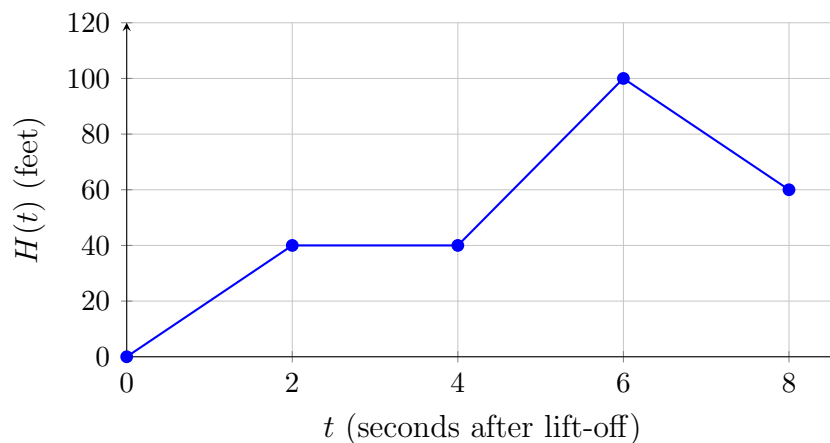
Function Notation, Domain, and Range

Evaluating a rule, finding the input for a given output, reading values from a table or a graph, and describing domain and range in context

Before you start.

- $f(3)$ is an instruction: put 3 in for the input and simplify. The answer is the *output*.
- “Find x when $f(x) = 21$ ” runs the same rule backwards: set the rule equal to 21 and solve.
- On a graph, a point (a, b) says $f(a) = b$. In a table, the input column and the output column say the same thing.
- The **domain** is the set of inputs allowed; the **range** is the set of outputs they produce. A story can shrink both.

Graph B. *Used only by Problems 2 and 3. No other problem on this sheet refers to it.* The graph shows $H(t)$, the height in feet of a survey drone t seconds after lift-off. The marked points are the readings the flight recorder stored.



1. Let $f(x) = 4x - 7$. Find $f(3)$.

Answer: _____

- 2.** Use **Graph B**. Find $H(4)$ — the drone's height 4 seconds after lift-off.

Answer: _____ ft

- 3.** Use **Graph B**. How much higher is the drone at $t = 6$ than it is at $t = 2$? Compute $H(6) - H(2)$.

Answer: _____ ft

- 4.** Let $g(x) = x^2 - 5x + 2$. Find $g(-2)$.

Answer: _____

- 5.** Use the same rule $f(x) = 4x - 7$ from Problem 1. Find the value of x for which $f(x) = 21$.

Answer: _____

- 6. Table A.** A rental shop charges $C(x)$ dollars to rent a kayak for x hours. The cost goes up by the same amount for each extra hour.

x (hours)	1	2	3	4
$C(x)$ (dollars)	22	34	46	58

- (a) By how many dollars does the cost rise for each extra hour?

Answer: _____

- (b) Write a rule for $C(x)$, then use it to find $C(6)$.

Answer: _____

- 7.** A tank holds $V(t) = 480 - 24t$ gallons of water t minutes after the drain is opened, and the drain runs until the tank is empty. What is the largest value of t for which this rule still describes the tank?

Answer: _____ min

8. Use $V(t) = 480 - 24t$ from Problem 7 again. After how many minutes does the tank hold 300 gallons?

Answer: _____ min

9. Use $V(t) = 480 - 24t$ once more. Find $V(9)$.

Answer: _____ gallons

10. **Find and fix the mistake.** Looking at **Table A** in Problem 6, a student wrote the rule $C(x) = 22x$ and then reported that renting a kayak for 5 hours costs 110 dollars.

Explain in one sentence what the student's rule ignores, then give the correct cost (in dollars) of a 5-hour rental.

Answer: _____

11. A theatre sells tickets to one show for 9 dollars each, and no single order may be larger than 8 tickets. The cost of an order of n tickets is $T(n) = 9n$ dollars.

(a) What is the largest value in the range of T for this situation (in dollars)?

Answer: _____

(b) Explain why the domain here is only the whole numbers from 0 to 8, even though the rule $T(n) = 9n$ by itself would accept any number.

12. Synthesis. You now have the same idea in three forms: the drone height on **Graph B**, the tank rule $V(t) = 480 - 24t$ from Problem 7, and the kayak costs in **Table A**.

Write a short paragraph that does three things:

- state a realistic domain and a realistic range for the drone function and for the tank function;
- say which form — rule, table, or graph — made each of those easier to see;
- explain why that form made it easier.